Leading Through Transition: Evaluating Engineering Transformation in a Scaling Organization

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Contents

Abstract

This paper examines the transformation of XYZ's engineering business unit during the company's transition from a startup to an SME. Faced with technical debt, delivery delays, and organizational stress, XYZ launched a major restructuring to increase throughput at its operational bottleneck — the engineering department. The new CTO implemented a task-aligned strategy emphasizing process redesign, modern engineering practices, and expert coaching. Using Lewin's Force Field Analysis and Bridges' Transition Model, the study evaluates how the organization managed both structural change and the associated human transition. The findings show that XYZ's success stemmed from removing restraining forces — such as complexity, mistrust, and skill gaps — rather than from intensifying pressure for change. Although the early phase of "letting go" was insufficiently supported, strong leadership, communication, and quick wins helped guide teams through the neutral zone toward full adoption. The transformation stabilized operations, improved collaboration, and established the foundation for sustained organizational learning.

Introduction

XYZ is a tech company that offers services through an online platform that allows contractors to upload their legal documents. The enterprise is transitioning from a small startup to an SME, reporting double-digit growth over the past six years.

This organic growth led to adverse emergent properties that inhibited the company's main value drivers, including the inability to deliver new features, missed market opportunities, customer dissatisfaction, and increased stress and anxiety within the organization.

XYZ's systemic analysis revealed that the engineering department was the organization's bottleneck. To increase the throughput of this constraint, the company initiated a revitalization effort primarily focused on this business unit. The adjustments made in this function subsequently resulted in changes at other organizational levels.

Methodology

This study adopts a qualitative case study approach to evaluate XYZ's organizational transformation during its transition from a startup to a medium-sized enterprise. The research focuses on the engineering business unit, identified as the company's operational bottleneck, and explores how structural and behavioural change unfolded within a high-growth context.

Data were collected from internal documentation, project retrospectives, and interviews with key stakeholders, including the Chief Technology Officer (CTO), engineering managers, and software developers involved in the transformation. These materials provided insights into both the formal restructuring process and the informal human dynamics that accompanied the change.

To analyze the change, two complementary theoretical models were applied: Lewin's Force Field Analysis and Bridges' Transition Model. Lewin's framework was used to assess the driving and restraining forces shaping organizational equilibrium and to examine how XYZ's strategy focused on removing obstacles rather than amplifying pressure for change. Bridges' model complemented this analysis by exploring the psychological dimension of transition, highlighting how individuals within the organization internalized and adapted to change over time.

This dual-lens methodology enabled a comprehensive evaluation of both the structural mechanisms and the human experiences that contributed to XYZ's transformation, aligning analytical rigour with actionable organizational insights.

Change Process and Implementation

Nature and Scope of the Transformation

XYZ's transformation was both structural and cultural in nature, initiated within the engineering department, the organization's operational bottleneck, and designed to restore performance, reliability, and alignment with business objectives.

At the structural level, the company appointed a new Chief Technology Officer (CTO) to spearhead the initiative. His first actions were decisive: he replaced the existing technical leadership, project managers, and senior developers with new hires and external experts who embodied the mindset he aimed to instill. To protect the teams from conflicting priorities and managerial interference, he restricted direct communication between upper management and engineering, thereby creating a clearer hierarchy of accountability.

Technologically, the new leadership introduced a completely different technical stack, deliberately breaking with legacy systems to foster a new organizational dynamic. Although this top-down decision triggered significant turnover, it served its intended purpose by accelerating the cultural reset and enabling the adoption of modern engineering practices.

In parallel with these structural and technological changes, external consultants collaborated with internal teams to implement state-of-the-art processes and tools. These included improved software quality assurance, iterative product design, and the introduction of self-organized, cross-functional teams supported by fit-for-purpose digital tools. Together, these interventions sought to replace outdated practices with an agile, quality-driven development culture.

The driving forces behind this transformation were both reactive and proactive. On one hand, chronic delivery delays and system instability pushed the company to consider rebuilding its core platform from the ground up. On the other hand, the prospect of entering the German market pulled the change forward, creating urgency for a more scalable and efficient technological foundation. The resulting transformation thus combined necessity with opportunity, balancing short-term remediation with long-term strategic ambition.

Strategic Approach to Change

XYZ's transformation followed a task-aligned strategy, as conceptualized by (Beer, Eisenstat, & Spector, 1990), emphasizing alignment among organizational structures, roles, and the specific work to be accomplished. The CTO began by redefining all roles and responsibilities within the engineering unit, replacing hierarchical control with accountability centred on task ownership. External experts were brought in as catalysts who modelled the desired practices and mentored internal staff. This approach enabled teams to progress organically through the classical group development stages (from forming and storming to norming) as they internalized new norms and workflows.

The development strategy also reflected a pragmatic prioritization. Indeed, initial efforts targeted peripheral features, allowing teams to experiment, learn, and refine their methods before tackling the company's core systems. This gradual convergence minimized operational risk while building technical and organizational maturity through iterative learning cycles.

After two years, all customers had successfully migrated to the new platform, and the legacy system was fully decommissioned. The engineering department stabilized with minimal subsequent turnover, signalling improved engagement and retention. As the external consultants completed their mandate, internal teams continued to develop new competencies autonomously, demonstrating that the change had evolved from a top-down intervention into a self-sustaining system of continuous improvement.

Managing Resistance and Building Commitment

The engineering department encountered barriers to change from both within and outside the business unit. Internally, most employees were initially **unfamiliar with the methods and technologies** they had to master. In addition to hiring expert consultants, the company overcame this barrier by fostering self-organization among teams, enabling the emergence of a "team of teams" (corresponding to Tuckman's performing stage).

Moreover, the **impatience and general lack of trust** expressed by the rest of the organization generated anxiety among employees within the engineering department. In response, XYZ adopted a strategy of releasing features early and often to achieve quick wins, systematically communicating these successes to reinforce understanding of the change and vision, and limiting commitments to what teams could realistically deliver.

Leadership and the Role of the Change Agent

Throughout the revitalization effort, the CTO relentlessly championed change and was accountable for its progress. As a change agent, he demonstrated a set of key capabilities, including strong communication, openness and adaptability, analytical thinking, leadership, and resilience.

He used these assets to openly and regularly communicate a **clearly articulated vision** of the desired future. Moreover, he remained positive yet realistic about the situation, celebrated small successes, and consistently collected **feedback**. In parallel, he surrounded himself with competent people (e.g., technical experts and leaders) to fill gaps in his expertise in specific areas.

However, he demonstrated limited listening skills and **empathy**. As a result, he did not effectively manage conflicts nor provide sufficient support to employees, particularly in interpersonal matters.

Assessment of the Organizational Change

The following section examines two essential facets of the revitalization effort: the **change process** (i.e., what is happening concretely) and the associated **transition** (i.e., how individuals come to accept change and leverage it over time).

The assessment draws on two complementary models. Lewin's Force Field Analysis is used to analyze the dynamics of the change process. Additionally, Bridge's Transition Model serves to evaluate the organization's transition management.

Lewin's Force Field Analysis

The Force Field Analysis provides a robust framework for planning change by identifying the forces influencing the system: the **driving forces**, which support the change, and the **restraining forces**, which act as obstacles (Spier, 1973).

It is a liberal instantiation of Newton's second law, which implies that the sum of all forces applied to a system in static **equilibrium** is zero. In this context, equilibrium refers to the point at which the organization's state stabilizes.

Initial Static Equilibrium

The following schema illustrates the force field before the revitalization effort.

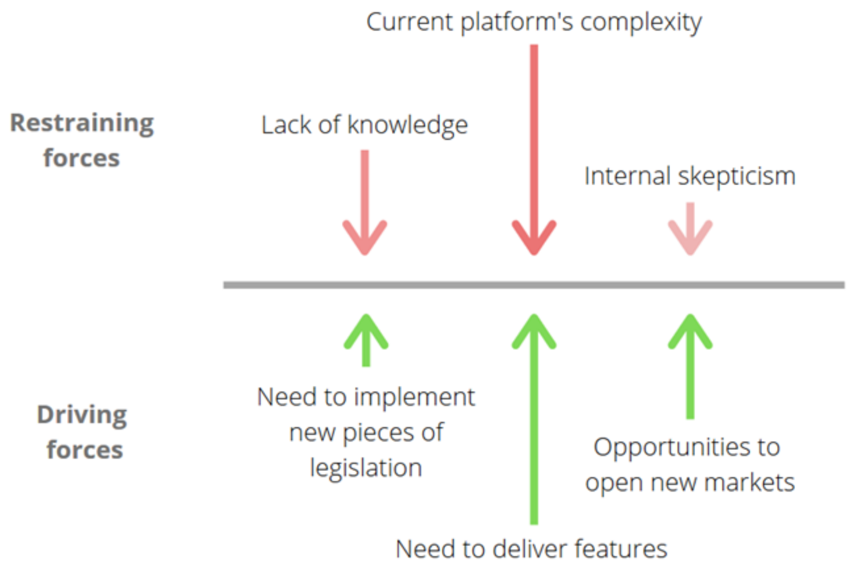

Figure 1: XYZ's force field before the change. The sum of all forces is zero, creating a static equilibrium (the status quo).

Moving the Equilibrium

XYZ decided to act on these forces to move the equilibrium toward the desired situation. The analysis below examines how these forces evolved as a result of the company's actions.

- **Opportunities to open new markets**: Emerging opportunities in Europe significantly strengthened this driving force.
- **Current platform's complexity**: XYZ reduced the pressure of this restraining force by building a leaner platform. As a result, teams were able to deliver new features without being constrained by the complexity of the existing platform.
- **Internal skepticism and lack of trust**: Fueled by the momentum generated through quick wins, this force ceased to function as an obstacle and evolved into a driver.
- **Lack of knowledge**: This force was transformed into a driver through the involvement of expert consultants, who transferred knowledge and new practices into the teams. In addition, the high turnover among long-tenured employees facilitated the adoption of these practices and reinforced this shift.

This analysis shows that XYZ's task-aligned strategy focused on **removing restraining forces** rather than merely adding new driving ones. This approach is more likely to result in **stable change**, as it eliminates forces that would otherwise push the system back toward previous behaviours — thereby reducing entropy (Spier, 1973; Beer, Eisenstat, & Spector, 1990).

Bridges’ Transition Model

The Transition Model focuses on understanding and managing transitions during change. According to this model, a transition is a *“three-phase process that people go through as they internalize […] the details of the new situation that the change brings”* (Bridges & Bridges, 2016). Ideally, the change process and transition occur concomitantly, but they can also happen on different timelines. It is also important to note that this process is not linear – the phases may overlap and recur over time (see Appendix 1).

The sections below evaluate XYZ’s experience across each phase of the model in relation to the actions it prescribes (Bridges & Bridges, 2016).

Phase 1 – Ending, Losing, Letting Go

People are often initially resistant to change because they must move away from a familiar situation in which they have developed habits. During this first phase, they should become open to change by acknowledging and accepting the end of the status quo. The figure below summarizes XYZ’s assessment for Phase 1.

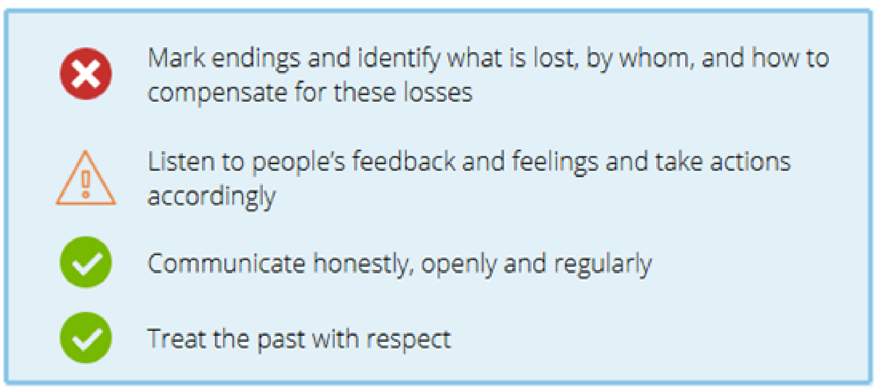

Figure 2: Assessment of XYZ during Phase 1.

According to (Beer & Nohria, 2000), change agents should balance Theory E (tasks and structure) and Theory O (people and emotions) to effect change efficiently. In this case, the CTO’s mindset was strongly biased toward Theory E, and he therefore listened primarily to task-related feedback. Consequently, employees grew resentful toward the organization, and many left as their work no longer aligned with their intrinsic motivations (EPM, 2018).

On the other hand, sharing the same information simultaneously with all employees prevented a “broken telephone” phenomenon and helped clarify why XYZ needed to change and what was in it for them. This approach also helped reduce the anxiety associated with uncertainty.

Finally, the CTO regularly acknowledged that past decisions had led the company to its current success, while clearly articulating why the status quo would not enable the organization to move forward.

Phase 2 – The Neutral Zone

The neutral zone is the most crucial phase of the transition, as it is a *"critical, ambiguous, uncertain time where leadership is highly required"* (Bridges & Bridges, 2016). The figure below summarizes XYZ's assessment for Phase 2.

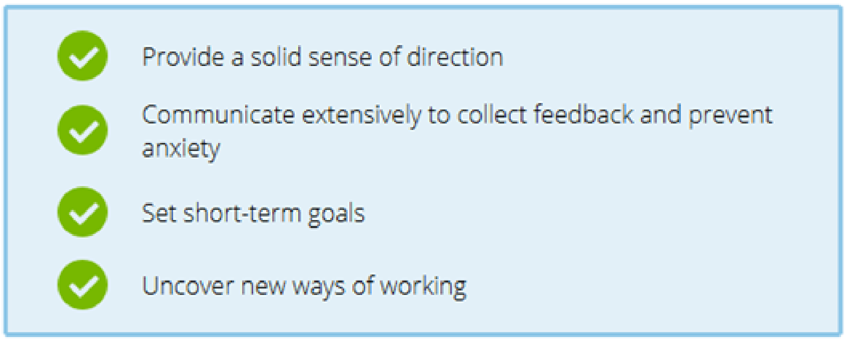

Figure 3: Assessment of XYZ during Phase 2.

This phase represents an important opportunity for creativity and innovation (Bridges & Bridges, 2016). By fostering self-organized teams, XYZ encouraged individuals to identify and solve problems autonomously.

Moreover, setting clear expectations prevented misaligned assumptions and aligned teams toward a shared direction, thereby increasing the likelihood of achieving the desired future state.

In addition, small and frequent releases enabled quick wins. The teams celebrated and communicated these incremental successes across the organization, reinforcing confidence and generating a sense of progress.

Finally, systematically collecting concerns enabled double-loop learning and allowed the CTO to adapt the course of action to address factors that could impede progress toward the desired future (Argyris, 1977).

Phase 3 – The New Beginning

In this final phase, individuals embrace the change and become open to learning the skills they will need to perform within the new context. The figure below summarizes XYZ's assessment for Phase 3.

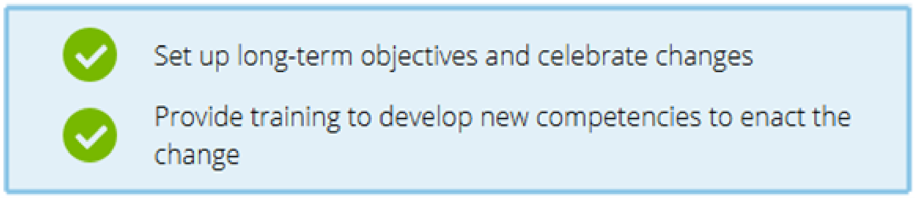

Figure 4: Assessment of XYZ during Phase 3.

The CTO's long-term goal was to replace the existing platform entirely with the new one. Although he did not explicitly articulate this objective early on, he began to communicate it more frequently as teams entered the transition phase. Each milestone reached served as an opportunity to celebrate progress and reinforce the team's commitment to this effort.

Furthermore, XYZ imposed new roles and responsibilities on employees to ensure long-lasting changes in behaviour (Beer, Eisenstat, & Spector, 1990). The company relied on expert consultants to provide coaching and support to individuals as they assumed their new responsibilities.

Overall Assessment

XYZ largely **overlooked** the first transition phase. However, this situation did not jeopardize the overall process, as most department employees were newcomers with limited attachment to prior practices. Moreover, **consistent and transparent communication** reduced anxiety and self-absorption, which are typically experienced during this phase (Bridges & Bridges, 2016).

On the other hand, the CTO demonstrated **strong leadership** in guiding teams through the subsequent phases — particularly the neutral zone. He also shielded employees from external disruptions, allowing them time and space to reorient.

Discussion

XYZ's transformation illustrates how structural and cultural renewal must evolve in parallel for organizational change to endure. The company's early-stage dynamics — a fast-moving but fragmented engineering culture — generated both innovation and instability. The CTO's task-aligned strategy effectively addressed these issues by redefining accountability, simplifying processes, and embedding learning mechanisms within teams. Through this approach, XYZ exemplified a shift from reactive adaptation to systemic alignment, in which structure, process, and culture reinforce one another.

Viewed through Lewin's Force Field Analysis, the CTO prioritized reducing restraining forces rather than amplifying driving forces. Instead of imposing pressure, he removed systemic barriers (e.g., legacy technology, overlapping roles, and conflicting directives) that constrained performance. This approach enabled the organization to unfreeze entrenched patterns organically, allowing new practices to gain legitimacy from the bottom up.

From a transitional perspective, the human dimension of change proved equally crucial. The CTO managed uncertainty by creating psychological distance between engineering teams and upper management, thereby providing the space necessary for experimentation and learning. Over time, this autonomy cultivated trust, ownership, and a collective sense of purpose, key elements for sustaining commitment once the external consultants departed.

Importantly, the process also demonstrated that cultural maturity develops incrementally through cycles of practice, reflection, and reinforcement. The progression from dependency on external expertise to internal self-sufficiency marks a critical inflection point in XYZ's organizational learning. By institutionalizing continuous improvement, the company transformed not only its systems but its capacity for sustained renewal.

Ultimately, XYZ's experience underscores that lasting change in high-growth organizations stems from strategic coherence and psychological alignment: the integration of formal redesign with human adaptation, through which operational excellence becomes a sustainable, self-reinforcing culture.

Conclusion

XYZ successfully implemented significant changes across its engineering business unit to address critical challenges.

The assessment of the company's change management using Lewin's Force Field Analysis and Bridges' Transition Model revealed that XYZ's task-aligned strategy effectively removed impediments to change. Moreover, the CTO's strong leadership guided the teams through the transition, even though his efforts during the initial mourning phase were insufficient.

These actions enabled durable change and generated lasting organizational benefits.

Appendices

Appendix 1. The Three Phases of Transition

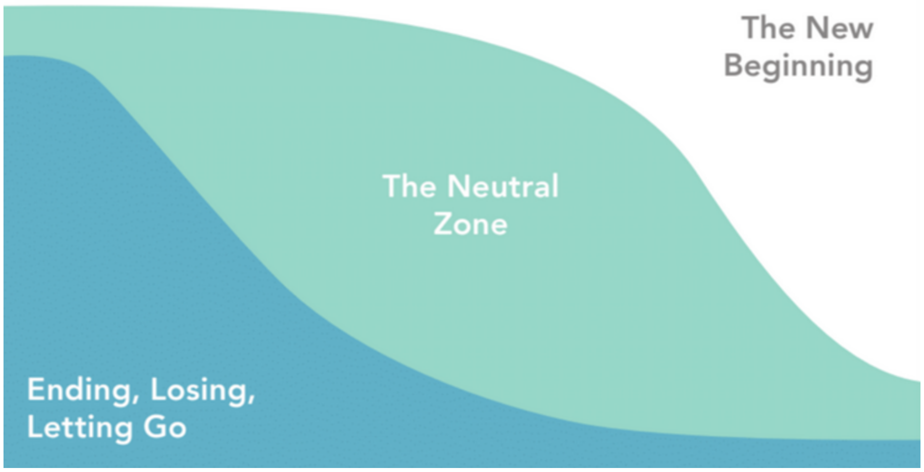

Figure 5: Bridges' three phases of transition. Phase 1 is more prominent at the beginning, and Phase 3 predominates toward the end. Source: letterpress.se

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